

Based on the provided transcript, here is the granular breakdown of exact requirements, architectural constraints, minor operational caveats, and explicit domain expectations for **Problem Statement 1 (Intelligent Conversational AI for KSP Crime Database)**.

1. Structural Requirements & Objectives

- **The Ultimate Objective:** The winning solution is not an academic exercise; it will be **taken all the way to production deployment** to actively empower investigators and senior law enforcement officers.
- **Primary Core Metric:** Drastically reduce investigation latency—specifically the time elapsed from the initial logging of an FIR (First Information Report) to the final step of putting an offender in jail.
- **Target Audience:** Exclusively built for police investigators, senior department officials, and state policymakers. It is strictly an **internal administrative system, not a public citizen-facing utility**.
- **Scale Target:** The system data architecture must be built and stress-tested to run robustly using a dataset scale of **100,000 to 200,000 cases (1 to 2 Lakh entries)** spanning up to 15–16 years of historical timelines across 1,100 police stations.

2. Technical & Infrastructure Constraints

- **Mandatory Cloud Platform:** The entire application and all its backend components **must be deployed natively on Zoho Catalyst**. Code running on alternative cloud vendors or external infrastructures is non-compliant.
- **Database & Schema Access:** No real-world production data will be distributed due to confidentiality constraints. The KSP team will provide the **master table structure, database schema, and column headers**.
- **Synthetic Data Generation:** Teams are required to populate the provided schemas with their own synthetic test data (1–2 Lakh entries) to evaluate performance and latencies.
- **Designations & Hierarchy Tree:** The system must implement a strict role-based access control matching the structural designations provided by KSP:
 - *Ranks:* Director General of Police (DGP), Inspector General of Police (IGP), Deputy Inspector General (DIG), Superintendent of Police (SP).
 - *Administrative Levels:* Chief Office, SP Office, Subdivisions, Local Police Stations.

3. Agentic & Architectural Expectations

- **Transition from Static Q&A to Agentic AI:** The panel emphasized that simple keyword or basic semantic Q&A chatbots are insufficient. The architecture must incorporate **Agentic AI**—utilizing AI agents capable of autonomous decision-making, stateful workflow execution, and complex task decomposition.
- **Advanced Relational Capability:** The agent must handle queries tracking complex network dynamics. It should automatically resolve linkages to discover if an individual suspect is tied across seemingly isolated multi-jurisdictional crimes.
- **Context Retention:** The LLM must preserve explicit conversation history locally, allowing

deep, multi-turn follow-up inquiries regarding a case, location, or MO without requiring the user to restate the operational baseline context.

- **Language & Modality:** Full stateful support for **text and voice interaction** natively in both English and Kannada from day one.
- **Local Session Logging:** Built-in programmatic functionality to compile and download conversation histories locally in standard **PDF format**.

4. Minor Observations, Edge Cases, & Warnings

- **LLM Choice Guardrails:** Explicit warning from the panel: **Do not attempt to pre-train or build an LLM from scratch**. It is an expensive and impractical task for this timeline. Instead, implement a foundational open-source model with open weights, host it on the allowed infrastructure, and fine-tune/ground it via Advanced RAG pipelines using the synthetic data.
- **Production Stability vs. Throwaway Code:** Solutions will be rigorously evaluated by senior cloud architects and domain experts. Codebases optimized for only a handful of users that break down under an increased multi-user load (e.g., scaling up from 10 to 100 concurrent sessions) will be penalized. Capacity planning, minimal latency, and horizontal scalability must be considered during design.
- **Outlier & Edge Case Tuning:** Mentors (domain experts like Mahesh and Balaji) will actively guide teams during the refinement phase to introduce regional anomalies, outliers, and edge cases to further harden the trained AI models.
- **Exclusion of Data Compliance Risks:** Dr. Pranamanti confirmed that since the distributed evaluation datasets are purely anonymized, artificial synthetic records, the regulatory constraints of the Digital Personal Data Protection (DPDP) Act do not apply to your development or testing sandbox phases.

Reference link : [▶ Datathon 2026: Introduction & Problem Statements Explainer Session.](#)